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The Impact of Artificial Intelligence in Advertising: An Experimental Evaluation of AI Video Generators

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**THE IMPACT OF ARTIFICIAL INTELLIGENCE IN ADVERTISING:
AN EXPERIMENTAL EVALUATION OF AI VIDEO GENERATORS**

A Thesis Submitted in Partial Fulfillment of the
Requirements for the Designation University Honors

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Introduction

Artificial intelligence (AI) has rapidly transformed the landscape of modern advertising, reshaping how agencies conceptualize, create, and deliver campaigns (Gujar and Panyam 51). Today, advertising agencies and standalone advertisers increasingly integrate AI tools to automate tasks, generate creative content, and analyze consumer behavior with precision. These technologies have introduced significant strengths, including enhanced creativity, efficiency, and time-saving capabilities, that allow marketers to produce highly targeted and engaging campaigns. However, the growing use of AI also raises questions about critical weaknesses, including issues of authenticity, bias, and ethical safety concerns in AI-generated content. Existing research has explored the general applications of AI in advertising and its implications for marketing strategy, yet gaps remain regarding how specific AI tools influence creative decision-making and efficiency in the process, particularly within video advertising.

To address this gap, this thesis evaluates the impact of artificial intelligence on advertising by conducting an experiment comparing the process and outcomes of three AI video generators in creating advertisements, using a standardized rubric. This project assesses the efficiency, quality, and creative potential of AI-generated advertising compared to traditional expectations, while identifying strengths and limitations in current generative AI tools. By focusing on the production process and final outputs of these AI systems, this research aims to provide a deeper understanding of how AI is shaping the future of video advertising and what its continued integration may mean for marketers, creators, and audiences.

Literature Review

How AI Tools Are Being Used in Advertising

The impact of AI on advertising is undeniable, with forecasts showing that 90% of ads will be created using AI in 2025 (Tur-Viñes et al. 84). AI is being used for a variety of tasks at ad agencies. Many ad agencies have fully implemented AI into their creative process through copywriting, campaign personalization, and content generation, to name a few (Gujar and Panyam 52). Other agencies have been hesitant to integrate the generative side of AI into their operations due to both the potential risks and the time investment required to train AI systems for the complex tasks needed. However, in those agencies, AI is still valued for its administrative tasks such as meeting planning, automation, and data analysis (Leszczynski et al. 2). The implementation of AI in marketing and advertising continues to grow in both the front-end creative development and the back-end data management processes. AI immediately established a strong presence in campaign management and data analysis to push forward campaigns. More recently, brands have shifted to utilizing and exploring the generative capabilities of artificial intelligence directly in the creative process. Brands such as Heinz and Coca-Cola have used generative AI openly in their advertisements, sparking both praise and controversy. Coca-Cola's generative holiday video advertisement ignited conversation over the integration of AI in ways that support and amplify human creativity, rather than allowing it to operate as an autonomous creator (Zargar 1). Burger King took a different approach by creating the "Million Dollar Whopper Contest," encouraging customers to build their own Whopper creation using generative AI and sharing it on social media. This campaign utilized AI to accelerate Burger King's digital adoption, noting that digital sales continue to increase by 40% every year (Littman 1). This application of AI shows how businesses and agencies can engage customers in interactive, participatory experiences with their customized AI tools. The range of uses for generative AI continues to expand and offer a wide horizon of opportunities for creative

innovation within advertising. As new AI applications develop, such as the most recent public rollout of Sora 2, the capabilities of artificial intelligence become stronger, as well as the ethical issues surrounding the use of artificial intelligence for generative purposes.

Strengths of AI Tools in Advertising

Artificial intelligence in advertising has reshaped the way ads are made. Agencies have been utilizing AI, particularly to aid in creativity, efficiency, and personalization (CE Noticias Financieras Editors 2). Specifically, generative AI has boosted engagement and conversion rates. Generative AI systems trained on diverse data sets produce unique content that catches the attention of consumers and yields results. Brands are now able to produce content at much higher rates than with just human power and easily tailor it toward their target audience, following purchasing behavior and preferences from available consumer data. Achieving these results using an AI system effectively cuts down on time and human power, which reduces costs exponentially (Gujar and Panyam 52). The automation of advertising processes and the effective rate at which AI is able to adapt strategies to increase campaign performance outdo what a human team would be capable of in the same amount of time and with the same budget (Ioniță et al. 41). Artificial intelligence's ability to produce unique, creative, and personalized results while maintaining efficiency and reducing costs has reshaped the way advertising content is conceptualized, produced, and delivered. These strengths, however, do not diminish the weaknesses apparent in artificial intelligence.

Weaknesses of AI Tools in Advertising

AI tools in advertising also present significant challenges, particularly regarding ethical concerns, dependence on data, and content quality. With the use of algorithms comes questions surrounding personal data and privacy risks. The use of personal data to create personalized ads raises concerns about disclosures, permissions, and confidentiality. Along with

this, the heavy reliance on data creates vulnerabilities for AI. Bias is prevalent because of the data AI systems are fed to produce their algorithms. Although practical methods can be put in place to avoid and reduce bias, this requires a constant evaluation of outputs and an ever-changing algorithm to ensure fair results (Ioniță et al. 42). Another weakness is the quality of the content created. Generative AI systems can produce a variety of results based on the software. Although these systems continue to develop, issues still occur with unnatural movements, unrealistic physics, and incorrect anatomy (Tur-Viñes et al. 84). For brands and agencies, the quality of the content is a major part of their final result, and it can make or break the success of an advertisement. Collectively, each of these limitations indicates the need for human intervention to ensure the responsible use of AI within advertising.

Video Advertisements

As AI continues to be utilized in the advertising industry, it is important to note the specific uses and the impact they have. Video advertising has emerged as one of the most dominant and impactful forms of digital marketing, with 90% of businesses implementing video advertising to help their customers understand their products and services and to boost brand recognition (Symon 1). Because of the major use of video advertisements and AI's growing relevance in creating and editing videos, video advertisements serve as a compelling area to focus on the growing relevance and practical implementation of generative AI.

How This Study Differs

While many current studies focus on consumer perception of outputs of generative AI and the generalized weaknesses and strengths of generative AI, this study will focus on the creative production process and the outputs of three specific AI tools - InVideo, Veed, and Pixlr - that focus on script-to-video generative works. Most prior work examines how AI influences marketing strategies or audience engagement rather than comparing the practical outcomes of

different AI systems. This thesis distinguishes itself by experimentally analyzing three AI video generators in different price ranges and evaluating their advertising process using a standardized rubric. The study will then go further to evaluate the consumer perception of the individual outputs of the AI video generators. While it is not the focus, the consumer perception of these tools is still a very important factor in evaluating their effectiveness. The results will offer a clearer understanding of how generative AI tools function for specific creative tasks while highlighting their practical applications and current limitations. This project contributes a unique perspective by evaluating the creative and functional capabilities of these AI tools and comparing them to each other to help understand their practical potential and the ongoing challenges of implementing generative AI in advertising.

Research Questions

- How will InVideo rank compared to the competitor tools based on efficiency, ease of use, cost effectiveness, prompt accuracy, visual aesthetics, storytelling quality, and final output quality?
- How will VEED rank compared to the competitor tools based on efficiency, ease of use, cost effectiveness, prompt accuracy, visual aesthetics, storytelling quality, and final output quality?
- How will Pixlr rank compared to the competitor tools based on efficiency, ease of use, cost effectiveness, prompt accuracy, visual aesthetics, storytelling quality, and final output quality?
- How do audience perceptions of AI-generated video content compare to perceptions of human-created video content?
- Which video source (low-cost AI, mid-cost AI, high-cost AI, or human-created) do viewers prefer overall?

- Is there a relationship between the price tier of an AI video generator and viewers' perceptions of the final output?

Methodology

To evaluate the AI tools, each was asked to create an advertisement for the University of Northern Iowa modeled from a [published video advertisement posted by the University of Northern Iowa](#). The published video was analyzed, and an AI chatbot created a general overview of the advertisement to help the AI tools model a video after it. To support the findings in past research, when the video scene was created, collaboration with humans was applied when necessary to fix hallucinations and mistakes. The script was also tweaked with human intervention based on the initial results of the AI-generated videos. To reduce bias, specific guidelines, tools, and techniques were created, noted in the experiment, and applied uniformly to each tool's outputs based on the tools' skills. Each tool was then tested and ranked based on a rubric much like the one modeled below. Once a final product is created by each tool, a survey will be conducted among students at the University of Northern Iowa to receive feedback on the visual aesthetics, storytelling quality, and final output quality.

AI Tool	Efficiency	Ease of Use	Cost Effectiveness*	Prompt Accuracy
InVideo	-	-	-	-
VEED	-	-	-	-
Pixlr	-	-	-	-

Rank 1-5, 1 as worst, 5 as best. Experiment notes will be given for each ranking.

*Cost will be specified and compared based on a cost-to-benefit analysis

AI Experimentation

Invideo, VEED, and Pixlr were utilized in a one-month subscription period to test the ability of the different tiers of generative AI tools to create an ad. In most cases, AI video generators are provided with detailed prompts to guide the output toward a specific result. While this approach was partially utilized in this study to refine videos after the initial generation, the primary focus remained on evaluating the generators' creative capacity and ability to produce realistic content. To support this objective, each video generator was instead given a general overview of the human-created UNI video ad, allowing for greater creative freedom in the generated outputs. The following prompt was used:

“Please create a 15-second promotional spot for Cedar Falls, Iowa, positioned specifically toward students considering the University of Northern Iowa. The video should communicate that Cedar Falls is a welcoming small city, a place to feel at home while away from home, and a centrally located community within Iowa. It has a very upbeat, aspirational, warm, and welcoming vibe. It uses UNI's branding colors, including hex codes #4B116F and #FFCC00, as well as their logo. The visuals should rely on aerial Cedar Falls city footage, lifestyle shots, UNI campus imagery, and UNI students walking together. It blends experiences off campus with academic life on campus. It emphasizes belonging, opportunity, and momentum. It positions the institution as not just an educational choice but a transformative opportunity. There is no dialogue, however music, on-screen text, and visual storytelling will be utilized to help to push the key points. The main message is that this is a place where you will meet your people, experience new things, grow as a person, and feel a sense of belonging both in Cedar Falls and at the University of Northern Iowa.”

As anticipated, the results varied across the different video generators. Minor adjustments to the original prompt were made when necessary to elicit more appropriate

outputs. The following evaluations provide an in-depth analysis of the modifications made for each video generator, as well as a comprehensive comparison of overall scores used to rank the AI tools on efficiency, ease of use, cost effectiveness, prompt accuracy, and overall comparative ranking seen in Figure 1 - AI Comparison Chart. The chart focuses on the analysis of the functions during the process rather than consumers' opinions on the final product, which is analyzed through the survey.

AI Tool	Efficiency	Ease of Use	Cost Effectiveness	Prompt Accuracy	Overall Comparative Ranking
InVideo	3	4	5	4	1
VEED	3	3	4	4	2
Pixlr	4	2	3	2	3

Figure 1 - AI Comparison Chart

InVideo

InVideo offers four tiered plans: free, plus, max, and generative. The max tier was chosen at \$120/mo providing 400 credits/mo, access to all AI models, including Veo 3.1, Sora 2 and Kling 3, access to all AI workflows, access to AI video trends, 16 AI avatars and voice clones, 2x more concurrency than Plus, 100 GB storage, 200 iStock, and unlimited exports without watermarks. The interface allowed for a script-to-video format with additional file uploads. The script was input without any edits made to the original copy; however, the current UNI logo was uploaded in addition to the script. Five total edits were made after the first video was created.

The first version incorporated an overlay of both music and sounds from the generated clips that varied from scene to scene. To change this, the generator was asked to “make the music pop, upbeat, and exciting and mute all other sound clips.” The next edit was to add a text overlay to add depth to the advertisement, as the original human-made video had. The request read, “Add text overlay on the first two shots. Make it white, bold, and relate to the main themes of the video.” The overlay input “CEDAR FALLS: YOUR NEW HOME” and “COMMUNITY. CONNECTION. DAY ONE.” on the first and second shots, respectively, as seen in Appendix 2.2. Additionally, in the original video, a sign was featured on a campus building with UNI’s old logo. The generator was asked, “At 7 seconds, change the sign in the background to the updated UNI logo previously uploaded.” The result was a generated version of the UNI logo, excluding the “University of Northern Iowa” text and the slash misplaced within the letters of UNI, as seen in Appendix 2.1. The edit was requested as, “Make sure the slash in the uploaded logo doesn’t go through the letters UNI. Include the whole logo on the sign at 7 seconds. No edits to it.” After this instruction failed, one last instruction attempt was made: “Move the slash on the ‘UNI’ sign to after the ‘I’ instead of in the middle. Keep the ‘University of Northern Iowa’ at the bottom. Model it after the official UNI logo uploaded.” However, instead of changing the original sign, the AI generator changed the entire scene and updated the sign to what was asked for as well.

Each video was uploaded at a standard pace, administering a rank of 3 on the efficiency scale. Within the interface, edits could be made to specific sections of the generated video without regenerating the entire video. This was not available on either of the other two AI generator interfaces. However, the edits were not always properly carried out, which earned the ease of use category a 4. Although InVideo was the highest priced, the many tools available and capabilities compared to the other two generators rank the highest at 5 for cost effectiveness. InVideo was able to create a complete video ad with minor changes on the first

try. Problems only occurred during the section editing portions. The prompt accuracy was ranked as 4, and InVideo was ranked as #1 overall.

Veed

Veed has three different pricing tiers: Creator, Pro, and Studio. Pro, the middle pricing tier, was chosen at \$49/mo, providing 2,500 credits, multiple brand kits, 30 minutes of translations, unlimited videos in Gen-AI Studio, and unlimited AI image generations. The interface allowed for a script-to-video format. The original script was input with settings set to horizontal. The first video created an avatar speaking a script with a mix of AI-generated and stock B-roll videos and images. After a few setting changes to create a fully AI video and remove the AI avatar, the video was sent into the editing studio within the interface. The editing studio provided features such as text overlay, audio edits, image additions, subtitle changes, elements, and a brand kit. UNI's logo was added to the brand kit, as well as UNI's two main colors of purple and gold. The video created had many errors. The spelling was wrong in many places, and non-cohesive scenes were put together, disregarding UNI's branding. The script was inserted again to generate a new video with two sentences added to the end: "Ensure that everything is spelled correctly. Put emphasis on the University of Northern Iowa's branding, including the purple and gold colors and the logo in the brand kit." The result included UNI's branding and built in more cohesive and natural scenes that could be mistaken for UNI's campus. However, the spelling was not fixed. In post-production, using Veed's editing workspace, text/rectangle overlays were used to change "Mommunity" to "Community" and "University of Northen Inowa" to "University of Northern Iowa."

On the efficiency scale, Veed ranked a 3. Although the loading was in line with the other interfaces, there were many failed attempts. The generated video would load for a few seconds, and then buffer without any progress. The only way to fix this issue was to reinsert the script to restart the entire generation process. Ease of use was also ranked a 3. Although the editing

workspace was a useful tool that could change many settings and help fix mistakes, the inability to regenerate sections of the video made it more difficult to work with overall. The cost effectiveness was rated a 4. Although less was expected due to the lower price, the buffering and blatant spelling errors that couldn't be fixed easily brought it down from a 5. However, there were a lot of features in the editing workspace that were useful in the post-production steps. The prompt accuracy was also ranked as a 4. The script was not fully followed, specifically with the spelling errors, and the video created was 12 seconds long rather than the 15 seconds that was asked for. Overall, VEED ranked second overall on the content creation process side.

Pixlr

Pixlr offered two pricing tiers: Plus and Premium. The premium was chosen at \$9.99/mo. The tier offered unlimited saves, 1000 monthly AI credits, a private mode for AI generation, and a library of fonts, templates, elements, animations, and more. Within the AI video generator section of the interface, there are three different tiers of quality a generated video can be made as: fast, pro, and ultra. The ultra tier was chosen when generating the video, as it is the only tier with audio added. The first video showed low-quality generated videos, misspelled words, and added the hex codes of UNI's color branding as text rather than just inputting the color scheme into the video. The duration was also only 8 seconds long instead of the 15 seconds that were requested. During the second attempt, the following was added to the original script: "Make sure the spelling of the text is correct in every shot. Make sure it is 15 seconds long." The second video did not show much improvement. Although slightly more aesthetic and with fewer spelling errors, the branded color hex code remained as text, the video was still 8 seconds long, and the University of Northern Iowa was spelled incorrectly as "University of Nurtion Ijwa". In post-production outside of the interface, a color-branded rectangle was used to cover the hex code, and new text was inserted over the misspellings.

Pixlr was ranked last overall compared to the other tools used. However, Pixlr ranked well in efficiency at 4. Due to less time spent on quality and duration, the videos were produced quite quickly. The ease of use was ranked as a 2 because of the inability to change specific sections of the video, the lack of editing features in post-production, and the inability to upload a branding guide or pictures to guide the generator in the right direction. The cost effectiveness was rated a 3. Although it was very cheap, the quality of the output provided was even less than the value of the subscription. Lastly, prompt accuracy was rated a 2. Although the video provided a loose idea of what the script was going for, none of the edits requested were made, and the original results of the script alone were poorly executed.

Overall, from a production standpoint, InVideo proved that the price was worthwhile for an efficient, easy-to-use tool that creates quality AI-generated content compared to VEED and Pixlr. Upon completion of the videos, they were placed in a survey to be evaluated and compared to each other and the human-made ad. The next phase of analysis for these videos was to test the perceptions of viewers who had no part in the creative process. Within the survey, participants were not informed which advertisements were generated using AI, allowing them to independently assess each video.

Survey Results

To answer the viewers' perceptions section of the research questions, a survey was conducted through Qualtrics among mainly UNI students, with seventy-five responses received in a week. Eighteen of the seventy-five responses were omitted due to being incomplete. The survey, found in Appendix 1, consisted of a series of questions about the four videos, a comparative section, and a demographic section. 88% (n=50) of the respondents were between the ages of 18-24, 4% (n=2) were between 25-31, 2% (n=1) were between 39-45, and the remaining 7% (n=4) were 46+. 72% (n=41) of the respondents were female, and 28% (n=16) were male. To gain a basis of their understanding and use of AI, they were asked how often

they use AI and what their primary use is. Four respondents said they never use AI, seven respondents rarely use AI, seven use AI a few times a month, eighteen respondents use AI a few times a week, ten respondents use AI once a day, and eleven respondents use AI multiple times a day. The usage of AI, displayed through Figure 2 - The AI Usage Chart below, had academic work as the top use, followed closely by brainstorming ideas, and writing and editing. The other category was used to explain that the respondents do not use AI.

36 - What do you primarily use AI for? - Selected Choice

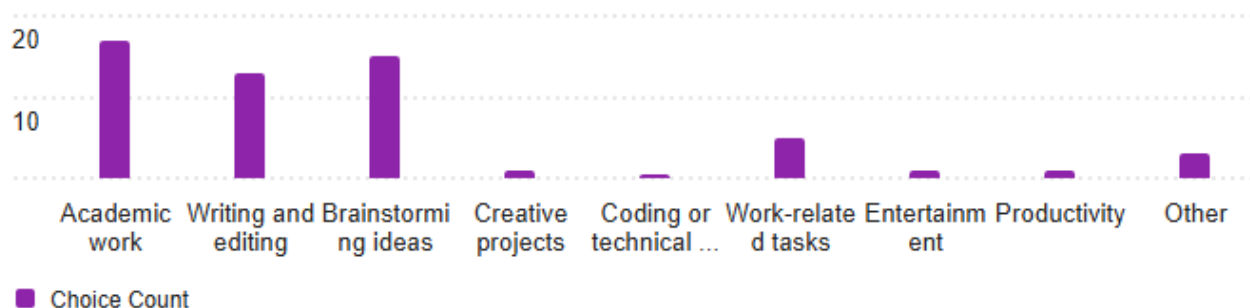


Figure 2 - AI Usage Chart

How will InVideo rank compared to the competitor tools based on visual aesthetics, message clarity, storytelling quality, and final output quality?

InVideo had a strong performance based on consumer perception compared to the other AI tools, with the strongest results in message clarity and overall output quality. In terms of visual aesthetics, 58% of respondents (n=33) rated the advertisement as somewhat or very appealing, while 18% (n=10) remained neutral, and 25% (n=14) noted some level of unappealingness with the video. Message clarity is a key strength, with 73% (n=42) of respondents indicating the message was somewhat or very clear, including 40% rating it very clear, while only 14% (n=8) reported any level of unclarity. 12% (n=7) of respondents remained neutral. Storytelling quality was the second biggest strength, with 63% (n=33) of responses between ratings of 6 and 10 and 26% at 8 or higher, indicating a generally effective but not

consistently high-level narrative. Overall output quality is concentrated in the upper range, with 63% of respondents rating the advertisement between 8 and 10 and only 16% at 6 or below, suggesting strong perceived final quality. These results indicate that InVideo performs well across all evaluated categories, with particularly strong communication clarity and solid overall output quality.

How will VEED rank compared to the competitor tools based on visual aesthetics, message clarity, storytelling quality, and final output quality?

VEED ranked below the human-created video and InVideo, but above Pixlr overall. In terms of visual aesthetics, 70% of respondents (n=40) rated the advertisement as somewhat or very appealing, including 51% selecting somewhat appealing and 19% very appealing, while 14% (n=8) reported some level of unappealing and 16% (n=9) remained neutral. Message clarity is relatively strong, with 70% of respondents indicating the message was somewhat or very clear, including 26% rating it very clear, though 19% reported some level of the message being unclear, and 11% remained neutral, suggesting some variability in communication effectiveness. Storytelling quality is distributed on the positive side, with 42% of responses (n=24) between ratings of 5 and 6 and 47% (n=27) between 7 and 10, indicating moderate narrative strength, as the remaining 11% fell between 3 and 4. Overall output quality is similarly centered in the mid to upper range, with 66% of respondents rating the advertisement between 6 and 8, and smaller portions reaching higher ratings, indicating generally acceptable performance. These results suggest that VEED delivers consistent but moderate outcomes across visual appeal, clarity, storytelling, and overall quality.

How will Pixlr rank compared to the competitor tools based on visual aesthetics, message clarity, storytelling quality, and final output quality?

Pixlr ranks as the lowest-performing option, with negative perceptions shown in all categories. 49% of respondents (n=28) rated the advertisement as somewhat or very unappealing, while only 26% (n=15) rated it as somewhat or very appealing, and 25% (n=14) remained neutral. The message clarity had varied responses, with 41% indicating the message was unclear, 19% remaining neutral, and 41% indicating a clear message. Storytelling quality is concentrated in the lower to mid-range, with 47% of responses between ratings of 3 and 5 and only 23% reaching 7 or higher, suggesting a less engaging narrative structure. Overall output quality reflects similar trends, with 56% of respondents rating the advertisement between 3 and 5, and only 22% rating it at 7 or above, indicating limited high-end performance. These results suggest that Pixlr struggles to deliver strong visual, narrative, and overall output quality based on audience perception.

How do audience perceptions of AI-generated video content compare to perceptions of human-created video content?

The human-made video demonstrates consistently strong performance across all evaluated categories, with survey results indicating high levels of audience approval, ranking it the top video overall. In terms of visual aesthetics, 93% (n=52) of respondents rated the advertisement as somewhat or very appealing, including 59% selecting very appealing, while only 4% (n=2) indicated any level of unappealing visuals, showing a clear preference for the visual quality. Message clarity is similarly strong, with 96% (n=54) of respondents indicating the message was somewhat or very clear and 71% rating it very clear, leaving only 4% neutral and no respondents reporting confusion. Storytelling quality is concentrated in the upper range, with 69% (n=38) of responses between ratings of 8 and 10 and only 13% at 6 or below, indicating a well-developed and engaging narrative. Overall output quality follows this same pattern, with

82% (n=46) of respondents rating the advertisement between 8 and 10, and nearly half of all responses at 9 or 10, demonstrating strong perceived quality. These results indicate that the human-made video performs at a high level across visual execution, communication clarity, narrative strength, and overall output.

Is there a relationship between the price tier of an AI video generator and viewers' perceptions of the final output?

The relationship between advertisement type and perceived use of artificial intelligence was evaluated using a chi-square test of independence at a significance level of 0.05. Survey responses were grouped into three perceived categories: not AI, neutral, and AI. The test produced a chi-square statistic of 96.2 with 6 degrees of freedom, resulting in a p-value less than 0.0001. Because the p-value is significantly below 0.05, the null hypothesis is rejected, indicating that perception of AI usage is strongly dependent on the tool used to create the advertisement. There were clear differences across tools, with the human-made advertisement predominantly identified as not AI, Pixlr overwhelmingly perceived as AI-generated, and both InVideo and VEED falling between these extremes. These results demonstrate that viewers are able to distinguish between content sources at a statistically significant level, suggesting that differences in output characteristics influence perceptions of AI involvement.

However, when respondents were directly asked about how suspecting any of these ads were AI affected the way they viewed the advertisements, 72% of respondents said it had a negative impact on their perception of the videos. The data also shows that the human-made video was viewed as preferred, more creative, and more authentic, and they were more likely to respond to it as a student, with 70%, 65%, 68%, and 65% demonstrated, respectively. This shows that although viewers are able to distinguish between the content sources with higher quality AI involvement perceived as less negative, the overall preference of viewers, without

being distinguished, is human-made ads. Furthermore, 88% of respondents noted that AI-generated advertisements feel less personal than human-made advertisements, with the remaining 7% staying neutral on the subject.

Overall AI Trends Perceptions

Survey participants were also asked about general trends of AI usage, specifically in the advertising industry. Survey results indicate that participants hold a cautious perspective toward the use of artificial intelligence in advertising, characterized by skepticism toward its expansion, concern about its implications, and recognition of its inevitability. A majority of respondents expressed resistance to increased AI usage, with 66% indicating disagreement with the statement that AI should be used more frequently in advertising, including 26% who strongly disagreed and 40% who somewhat disagreed, while only 14% expressed agreement. This resistance is further supported by perceptions of reduced personalization, as 88% of participants agreed that AI-generated advertisements feel less personal than human-made content, including 58% who strongly agreed. Despite these concerns, respondents still acknowledge the significant role AI will play in the future of the industry, with 87% agreeing that AI will significantly change advertising, including 54% who strongly agreed. Interestingly, this skepticism does not translate into outright rejection of AI-generated content, as comfort levels remain relatively balanced, with 26% of respondents indicating some level of comfort, 42% expressing discomfort, and 32% remaining neutral. Additionally, 72% of participants reported that suspecting an advertisement was AI-generated negatively affected their perception, reinforcing the presence of bias even when content is consumed. Collectively, these findings suggest that while participants are aware of and expect the growing influence of AI in advertising, they do not necessarily support its expansion, citing concerns related to authenticity, personalization, and overall perception.

Limitations and Recommendations

The survey was conducted in a short amount of time at the University of Northern Iowa, with constraints during the distribution process. Because of this, a smaller convenience sample size was represented. The sample size consisted of 72% female and only 28% male, indicating a notable imbalance in gender representation. Along these lines, 87.7% of respondents were in the age group of 18- to 24-year-olds. This study was therefore highly concentrated in a younger age group, and opinions reflect as such.

In the future, improvements to the research design would strengthen the generalizability of the results. First, the distribution of the survey would be expanded to include a larger, more diverse sample. The current dataset is heavily connected to the students at the University of Northern Iowa. A more representative sample could be achieved by increasing the number of participants and potentially using paid survey distribution methods to ensure broader demographic coverage across age, gender, and geographic location.

In addition, future studies should evaluate how familiarity with the content influences perception. Because the advertisements were centered around UNI's campus and given to respondents who recognize the location, the videos were most likely interpreted differently than those who do not know the campus. Conducting a controlled comparison between viewers who are familiar with the campus and those who are not would provide insight into whether contextual knowledge affects perceived authenticity, clarity, and overall quality.

Along with this, the selection of AI tools could be more refined to highlight the capabilities of AI rather than to evaluate the differences in perception and performance, based on user interfaces with limited features and varied pricing. Instead, the focus could be a direct assessment of AI video generation models such as Kling, VEO, and Sora. This approach would reduce the ability of consumers to identify AI-generated content and put less emphasis on the user interfaces of each platform based on pricing.

An additional area of focus for this study is to hone in on how the AI-generated content is prompted. This study intentionally emphasized the creative capabilities of AI by allowing the tools to generate advertisements with relatively open-ended prompts, rather than providing highly detailed or structured instructions. As a result, the evaluation focused more heavily on storytelling quality and narrative development, where the outputs often demonstrated limited depth and less cohesive structure. Future research could adopt a more controlled prompting approach by supplying highly specific and detailed inputs to better isolate the production quality of AI-generated content. This would allow for a clearer distinction between limitations in storytelling versus limitations in visual quality. Additionally, comparing how different AI generation models respond to both open-ended and highly detailed and directed prompts would provide deeper insight into whether insufficiencies in narrative depth are consistent across models or dependent on the underlying technology. This approach would help determine whether AI's current limitations are primarily creative or technical in nature.

Conclusion

This study evaluated the effectiveness of AI-generated advertisements in comparison to human-created content, with a focus on visual aesthetics, message clarity, storytelling quality, and overall output quality. The results indicate that while AI-generated tools are capable of producing acceptable and in some cases high-quality outputs, they do not yet consistently match the performance of human-created advertisements, particularly in areas requiring narrative depth and cohesive storytelling. The human-made video demonstrated the strongest performance across all evaluated categories, with high concentrations of responses in the upper rating ranges, reinforcing the continued advantage of human creativity in producing engaging and well-structured content.

This finding, along with the ranking of the AI tools from both a production and viewer perception standpoint, validated the anticipated results made before the study began. Among the AI tools evaluated, performance varied notably depending on the platform. InVideo demonstrated strong results in message clarity and overall output quality, suggesting that AI tools can effectively communicate information and produce polished final products. However, storytelling quality remained a limitation for all AI tools. VEED produced consistent but moderate results across all categories, while Pixlr showed weaker performance, particularly in visual appeal and overall quality. These findings suggest that while AI tools can replicate certain aspects of advertisement production, their effectiveness is highly dependent on the platform and its underlying capabilities.

Overall, this study demonstrates that the creativity of AI models is not yet sufficient compared to traditional content creation made by humans. While strengths in clarity and efficiency are evident, limitations in storytelling and consumer perception remain significant barriers. Consumers are also resistant to an increased use of AI in advertising and hold a negative perception. However, the discomfort of consuming AI is less of a concern at the moment, as the results are statistically insignificant. These findings suggest that while AI tools are advancing rapidly, both their technical capabilities and public perception present limitations that must be addressed when considering the future use of artificial intelligence in advertising.

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Appendix

1.1 Ad # 1 - "Cedar Falls - Home. On Day One."

<https://youtu.be/LqGFMcRzYRM>

1.2 Ad #2 - "Cedar Falls is the perfect place for your student to learn, grow and thrive"

<https://youtu.be/drrxhEFwXbY>

1.3 Ad #3 - "Find Your Place"

<https://youtu.be/1Mthvwn2olq>

1.4 Ad #4 - "Embrace Cedar Falls Community"

<https://youtu.be/nhm-dZd6COg>

2.1 UNI Old Logo Scene



2.2 Attempted Old Logo Revision



3.1 Survey

Impact of AI on Advertising

Start of Block: Video 1

View this video and answer the following questions - UNI Video 1

1 How would you rate the overall quality of this advertisement?

- ☐ 0 (0)
- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)
- ☐ 7 (7)
- ☐ 8 (8)
- ☐ 9 (9)
- ☐ 10 (10)

2 How visually appealing did you find this advertisement?

- ☐ Very unappealing (1)
- ☐ Somewhat unappealing (2)
- ☐ Neither appealing or unappealing (3)
- ☐ Somewhat appealing (4)
- ☐ Very Appealing (5)

3 How clear was the message of this advertisement?

- ☐ Very unclear (1)
- ☐ Somewhat unclear (2)
- ☐ neutral (3)
- ☐ Somewhat clear (4)
- ☐ Very clear (5)

4 How would you rate the quality of the storytelling in this ad?

- ☐ 0 (0)
- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)
- ☐ 7 (7)
- ☐ 8 (8)
- ☐ 9 (9)
- ☐ 10 (10)

5 Do you believe this advertisement was created using AI?

- ☐ Definitely not (1)
- ☐ Probably not (2)
- ☐ Might or might not (3)
- ☐ Probably yes (4)
- ☐ Definitely yes (5)

Start of Block: Video 2

Q14 View this video and answer the following questions - UNI Video 2

6 How would you rate the overall quality of this advertisement?

- ☐ 0 (0)
- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)
- ☐ 7 (7)
- ☐ 8 (8)
- ☐ 9 (9)
- ☐ 10 (10)

7 How visually appealing did you find this advertisement?

- ☐ Very unappealing (1)
- ☐ Somewhat unappealing (2)

- ☐ Neither appealing or unappealing (3)
- ☐ Somewhat appealing (4)
- ☐ Very Appealing (5)

8 How clear was the message of this advertisement?

- ☐ Very unclear (1)
- ☐ Somewhat unclear (2)
- ☐ neutral (3)
- ☐ Somewhat clear (4)
- ☐ Very clear (5)

9 How would you rate the quality of the storytelling in this ad?

- ☐ 0 (0)
- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)
- ☐ 7 (7)
- ☐ 8 (8)
- ☐ 9 (9)
- ☐ 10 (10)

10 Do you believe this advertisement was created using AI?

- ☐ Definitely not (1)

- ☐ Probably not (2)
- ☐ Might or might not (3)
- ☐ Probably yes (4)
- ☐ Definitely yes (5)

Start of Block: Video 3

Q21 View this video and answer the following questions - UNI Video 3

11 How would you rate the overall quality of this advertisement?

- ☐ 0 (0)
- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)
- ☐ 7 (7)
- ☐ 8 (8)
- ☐ 9 (9)
- ☐ 10 (10)

12 How visually appealing did you find this advertisement?

- ☐ Very unappealing (1)
- ☐ Somewhat unappealing (2)
- ☐ Neither appealing or unappealing (3)

- ☐ Somewhat appealing (4)
- ☐ Very Appealing (5)

13 How clear was the message of this advertisement?

- ☐ Very unclear (1)
- ☐ Somewhat unclear (2)
- ☐ neutral (3)
- ☐ Somewhat clear (4)
- ☐ Very clear (5)

14 How would you rate the quality of the storytelling in this ad?

- ☐ 0 (0)
- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)
- ☐ 7 (7)
- ☐ 8 (8)
- ☐ 9 (9)
- ☐ 10 (10)

15 Do you believe this advertisement was created using AI?

- ☐ Definitely not (1)
- ☐ Probably not (2)

- ☐ Might or might not (3)
- ☐ Probably yes (4)
- ☐ Definitely yes (5)

Start of Block: Video 4

Q28 View this video and answer the following questions - UNI Video 4

16 How would you rate the overall quality of this advertisement?

- ☐ 0 (0)
- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)
- ☐ 7 (7)
- ☐ 8 (8)
- ☐ 9 (9)
- ☐ 10 (10)

17 How visually appealing did you find this advertisement?

- ☐ Very unappealing (1)
- ☐ Somewhat unappealing (2)
- ☐ Neither appealing or unappealing (3)
- ☐ Somewhat appealing (4)

- ☐ Very Appealing (5)

18 How clear was the message of this advertisement?

- ☐ Very unclear (1)
- ☐ Somewhat unclear (2)
- ☐ neutral (3)
- ☐ Somewhat clear (4)
- ☐ Very clear (5)

19 How would you rate the quality of the storytelling in this ad?

- ☐ 0 (0)
- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)
- ☐ 7 (7)
- ☐ 8 (8)
- ☐ 9 (9)
- ☐ 10 (10)

20 Do you believe this advertisement was created using AI?

- ☐ Definitely not (1)
- ☐ Probably not (2)
- ☐ Might or might not (3)

- ☐ Probably yes (4)
- ☐ Definitely yes (5)

Start of Block: Comparative Questions

21 Which ad did you prefer overall?

- ☐ Ad #1 - "Cedar Falls - Home. On Day One." (1)
- ☐ Ad #2 - "Cedar Falls is the perfect place for your student to learn, grow and thrive" (2)
- ☐ Ad #3 - "Find Your Place" (3)
- ☐ Ad #4 - "Embrace Cedar Falls Community" (4)

22 Which ad felt more creative overall?

- ☐ Ad #1 - "Cedar Falls - Home. On Day One." (1)
- ☐ Ad #2 - "Cedar Falls is the perfect place for your student to learn, grow and thrive" (2)
- ☐ Ad #3 - "Find Your Place" (3)
- ☐ Ad #4 - "Embrace Cedar Falls Community" (4)

23 Which ad felt more authentic overall?

- ☐ Ad #1 - "Cedar Falls - Home. On Day One." (1)
- ☐ Ad #2 - "Cedar Falls is the perfect place for your student to learn, grow and thrive" (2)
- ☐ Ad #3 - "Find Your Place" (3)
- ☐ Ad #4 - "Embrace Cedar Falls Community" (4)

24 Which ad would you be more likely to respond to as a student?

- ☐ Ad # 1 - "Cedar Falls - Home. On Day One." (1)
- ☐ Ad #2 - "Cedar Falls is the perfect place for your student to learn, grow and thrive" (2)

- o Ad #3 - "Find Your Place" (3)
- o Ad #4 - "Embrace Cedar Falls Community" (4)

Answer the following with the degree you agree/disagree with the statements made.

25 AI should be used more frequently in advertising.

- o Strongly disagree (1)
- o Somewhat disagree (2)
- o Neither agree nor disagree (3)
- o Somewhat agree (4)
- o Strongly agree (5)

26 AI-generated ads feel less personal than human-made ads.

- o Strongly disagree (1)
- o Somewhat disagree (2)
- o Neither agree nor disagree (3)
- o Somewhat agree (4)
- o Strongly agree (5)

27 I am comfortable consuming ads created by AI.

- o Extremely uncomfortable (1)
- o Somewhat uncomfortable (2)
- o Neither comfortable nor uncomfortable (3)
- o Somewhat comfortable (4)
- o Extremely comfortable (5)

28 AI will significantly change the advertisement industry in the future.

- ☐ Strongly disagree (1)
- ☐ Somewhat disagree (2)
- ☐ Neither agree nor disagree (3)
- ☐ Somewhat agree (4)
- ☐ Strongly agree (5)

29 Did suspecting any of these ads were AI-generated affect how you viewed them?

- ☐ Yes, postively (1)
- ☐ Yes, negatively (2)
- ☐ No (3)
- ☐ I did not suspect any of these ads were AI-generated (4)

Display this question:

If Did suspecting any of these ads were AI-generated affect how you viewed them? = Yes,
postively

Or Did suspecting any of these ads were AI-generated affect how you viewed them? = Yes,
negatively

30 Which ad/ads did you suspect were AI-generated?

- ☐ Ad #1 - "Cedar Falls - Home. On Day One. (1)
- ☐ Ad #2 - "Cedar Falls is the perfect place for your student to learn, grow and thrive" (2)
- ☐ Ad #3 - "Find Your Place" (3)
- ☐ Ad #4 - "Embrace Cedar Falls Community" (4)

Display this question:

If If Which ad/ads did you suspect were AI-generated? q://QID55/SelectedChoicesCount Is Not Empty

31 How, specifically, did suspecting the ad(s) was/were AI-generated affect how you viewed it positively/negatively?

32 Do you have any concerns about AI being used in advertising? If so, please explain.

Start of Block: Demographics

33 What is your age?

- ☐ Under 18 (1)
- ☐ 18-24 (2)
- ☐ 25-31 (3)
- ☐ 32-38 (4)

- o 39-45 (5)
- o 46+ (6)

34 What is your gender?

- o Male (1)
- o Female (2)
- o Non-binary (3)
- o Other (4) _____

35 How often do you use AI?

- o Never (1)
- o Rarely (2)
- o A few times a month (3)
- o A few times a week (4)
- o Once a day (5)
- o Multiple times a day (6)

36 What do you primarily use AI for?

- o Academic work (1)
- o Writing and editing (2)
- o Brainstorming ideas (3)
- o Creative projects (4)
- o Coding or technical tasks (5)
- o Work-related tasks (6)
- o Entertainment (7)
- o Productivity (8)

- o Other (9) _____

End of Block: Demographics